Machine Learning

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Contents

I. Introduction

II. Machine Learning Basics

V. III. Linear Models

VI. IV. Neural Networks

VII. V. Support Vector Machines

III. VI. Decision Trees

IV. VII. Statistical Learning

VIII. Learning Theory

IX. Principles of Deep Learning

X. Ensemble Methods and Meta Learning

XI. Reinforcement Learning

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Objectives

- understand and explain the basic concepts of machine learning
- understand formalized concepts and methods

- be able to implement concepts and methods in the form of algorithms
- be able to sensibly select, adapt, and apply relevant methods
- be able to educate oneself

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Related Fields

1. Statistics [paradigms, models]
2. Mathematics

3. Information Retrieval [methods, algorithms]
4. Knowledge Processing
5. Heuristic Search

6. Decision Support Systems [applications]
7. Artificial Intelligence
8. Web Technology

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Literature

Machine Learning:

- C.M. Bishop. [Interview (2018 MS Research)]
  Pattern Recognition and Machine Learning
  2nd edition, Springer 2006.

- L. Breiman, J.H. Friedman, R.A. Olshen, C.J. Stone.
  Classification and Regression Trees
  CRC Press reprint, 1998.

- N. Cristianini, J. Shawe-Taylor.
  An Introduction to Support Vector Machines and Other Kernel-based Learning Methods
  Cambridge University Press, 2000.

- T. Hastie, R. Tibshirani, J. Friedman.
  The Elements of Statistical Learning
  2nd edition, Springer, 2009. statweb.stanford.edu/∼hastie/ElemStatLearn/

- T. Mitchell.
  Machine Learning
  1st edition, McGraw-Hill, 1997. www.cs.cmu.edu/∼tom/mlbook.html

- V. Vapnik.
  The Nature of Statistical Learning Theory
  2nd edition, Springer 2000.

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Software

Programming:

- Eclipse Foundation, Inc., Canada.
  Eclipse SDK
  Version 4.13. www.eclipse.org/downloads

Machine Learning:

- E. Frank, M. Hall, G. Holmes, M. Mayo, B. Pfahringer, T. Smith, I. Witten.
  Weka Machine Learning Project
  Version 3.8. www.cs.waikato.ac.nz/ml/weka

- *scikit-learn: Machine Learning in Python*
  Version 0.21 http://scikit-learn.org/stable/

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Software

Statistics:

- R Development Core Team.
  R
  Version 3.6. www.r-project.org

- E. Jones, T. Oliphant, P. Peterson and others.
  SciPy
  Version 1.17. www.scipy.org

- J.W. Eaton.
  GNU Octave
  Version 5.1. www.gnu.org/software/octave

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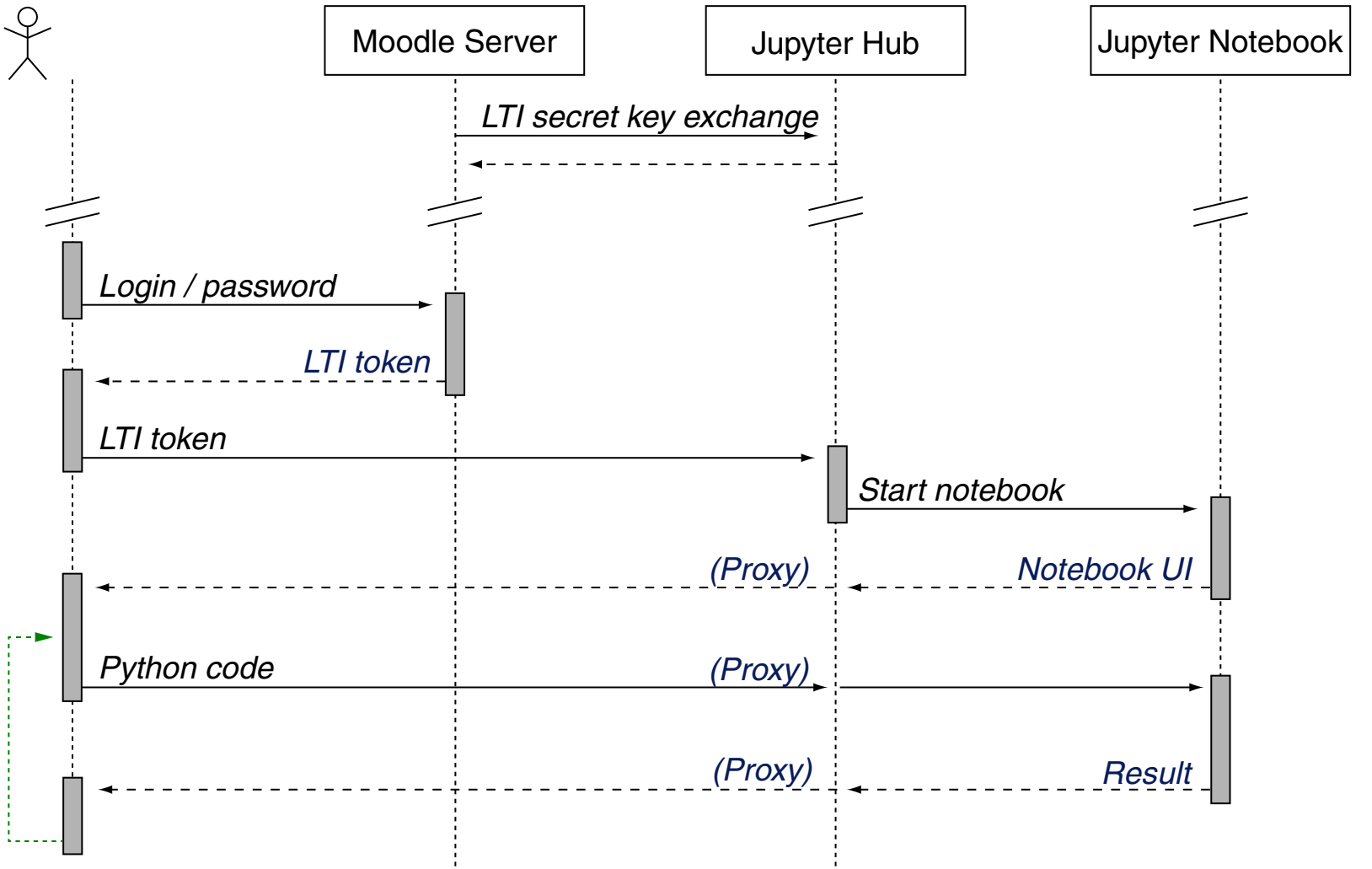
Software

Lab Class Setup



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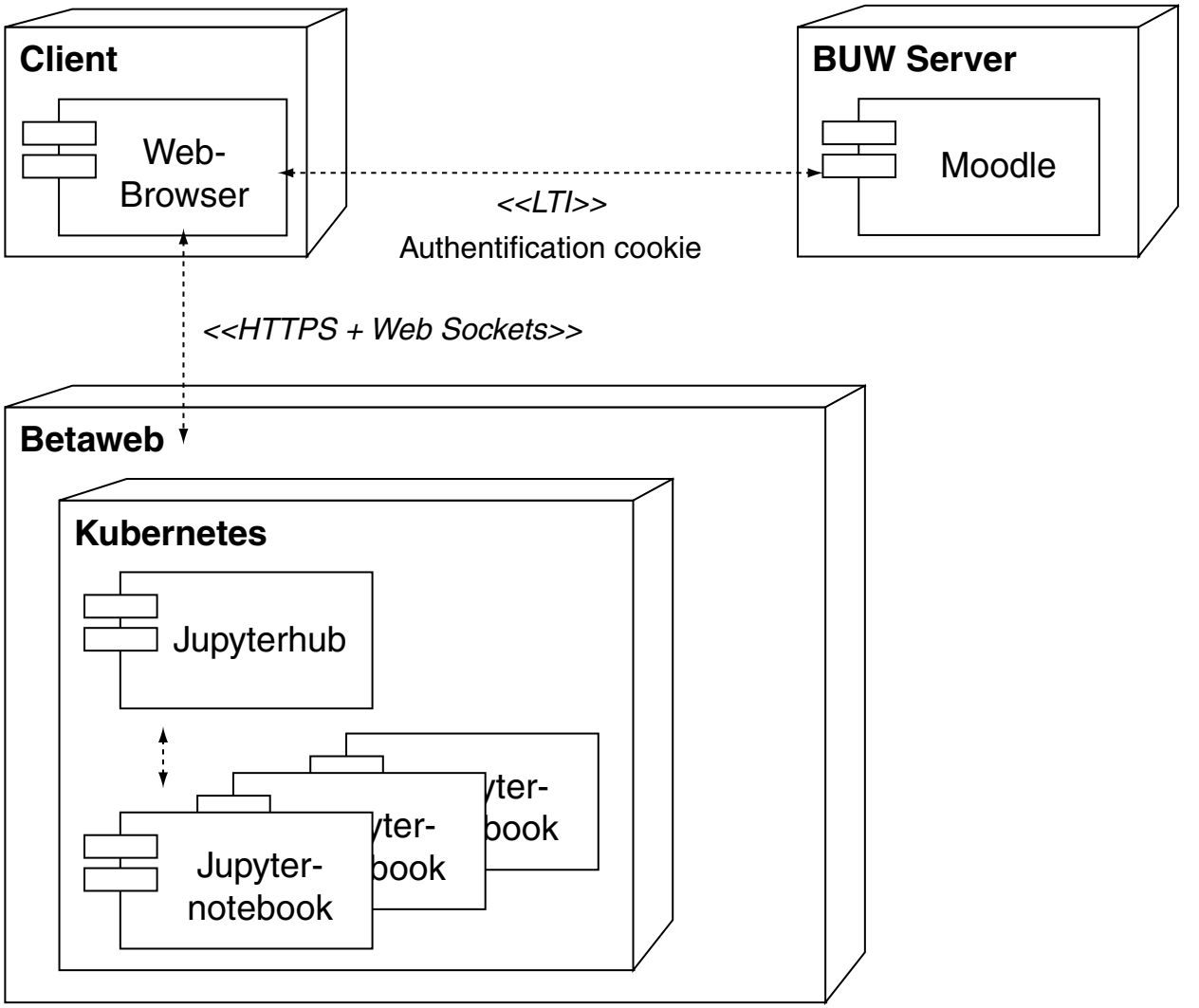
Software

Lab Class Setup (continued)



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